

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T) \times V \rightarrow \text{Summary}$

Day - 6

Loc 74

Scaled Dot Product Attention

Original paper : $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$

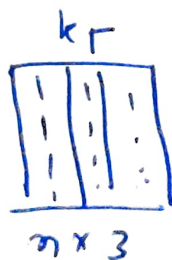
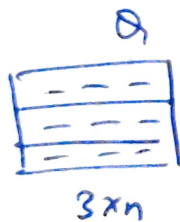
$d_k \rightarrow$ dimensionality of key vector.

for avoiding unstable gradients

Our case , $d_k = d_q = d_v = 3$

$\Rightarrow \text{softmax}\left(\frac{QK^T}{\sqrt{3}}\right) V$

Why? $\rightarrow$ Dot product's nature



$\Rightarrow$ 9 values $\Rightarrow$ 4
6

Low-dimensionality $\rightarrow$ dot product $\rightarrow$ low variance

$(1, 3) \times (3 \times 7) \times (7 \times 3) \Rightarrow 9 \text{ values} \rightarrow \text{variance high}$

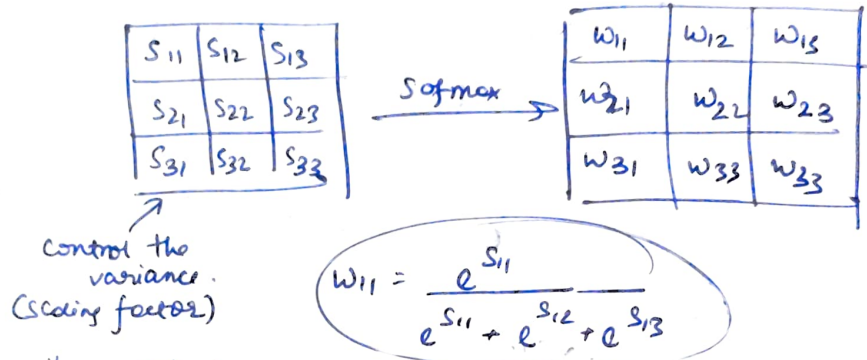
$(3 \times 3) \times (7 \times 3) \Rightarrow 9 \text{ values} \rightarrow \text{variance low}$

1) (1,2) (3,2) a
 2) (.,.) (.,.) b
 3 c
 4 d

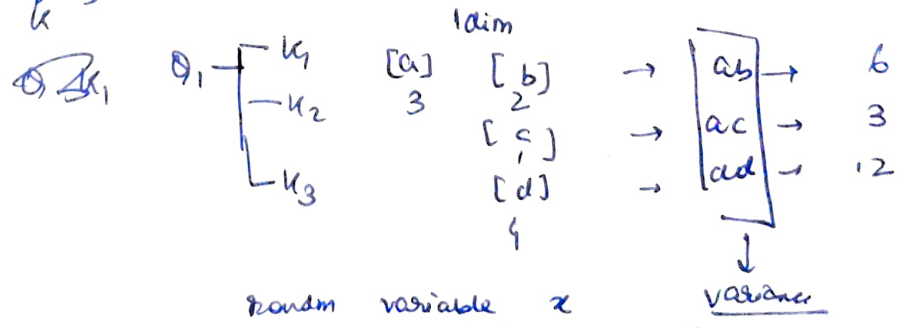
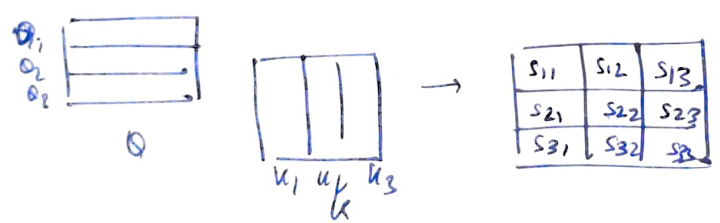
1) (1,2,3), (4,5,6)
 2
 3
 4
 σ_1^2 σ_2^2

$\sigma_2^2 > \sigma_1^2$

dim $\uparrow \Rightarrow$ variance $\uparrow$
 high variance is a problem



if $\sigma^2 \uparrow \Rightarrow w_{11} \uparrow, w_{12} \downarrow, w_{13} \downarrow$
 • some weights will be very high
 • small weights will be ignored
 • B VGB



2dim

$[a \ b] \ [c \ d] \rightarrow ac+bd$
 $[e \ f] \rightarrow ae+bf$
 $[g \ h] \rightarrow ag+bh$
 1? $\downarrow$
 $var(y)$

$var(y) > var(x)$
 'yes' No x
 $var(y) \approx 2 var(x)$

3 dim

$$\begin{aligned} [a \ b \ c] & [d \ e \ f] \rightarrow ad + be + cf \\ [g \ h \ i] & \rightarrow ag + bh + ci \\ [k \ l \ m] & \rightarrow dk + el + fm \end{aligned} \left. \vphantom{\begin{aligned} [a \ b \ c] \\ [g \ h \ i] \\ [k \ l \ m] \end{aligned}} \right\} \text{Var}(z)$$

$$\text{var}(z) \approx 3 \text{var}(x)$$

$$\Rightarrow \begin{aligned} 1 \text{ dim} & \rightarrow \text{var}(x) \\ 3 \text{ dim} & \rightarrow 3 \text{var}(x) \\ & \vdots \\ d \text{ dim} & \rightarrow d \text{var}(x) \end{aligned}$$

If you have a random variable x with a variance of $\text{var}(x)$ & you create a new variable y by scaling x with const c , so that $y = cx$,

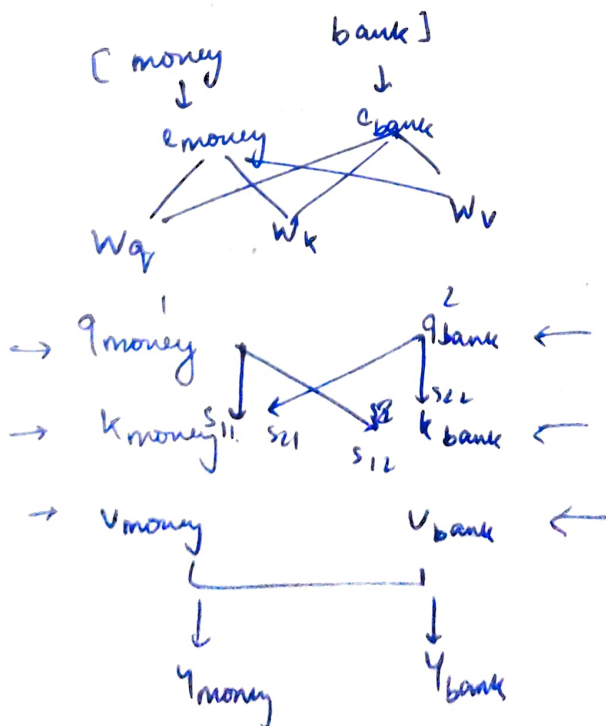
then $\text{var}(y) = c^2 \text{var}(x)$

$$\Rightarrow y = c \cdot x$$

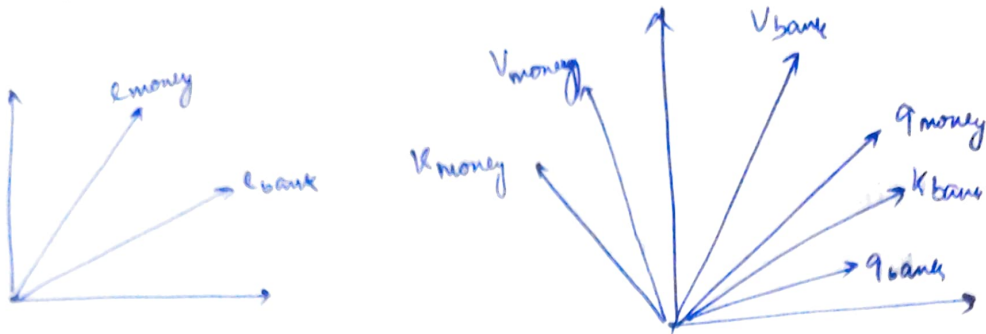
$$\therefore (x) = \frac{(y)}{\sqrt{d}} \text{ for } d \text{ dim. } \underline{\text{Key vector}}$$

Lec-75

Keep Attention Geometric Intuition

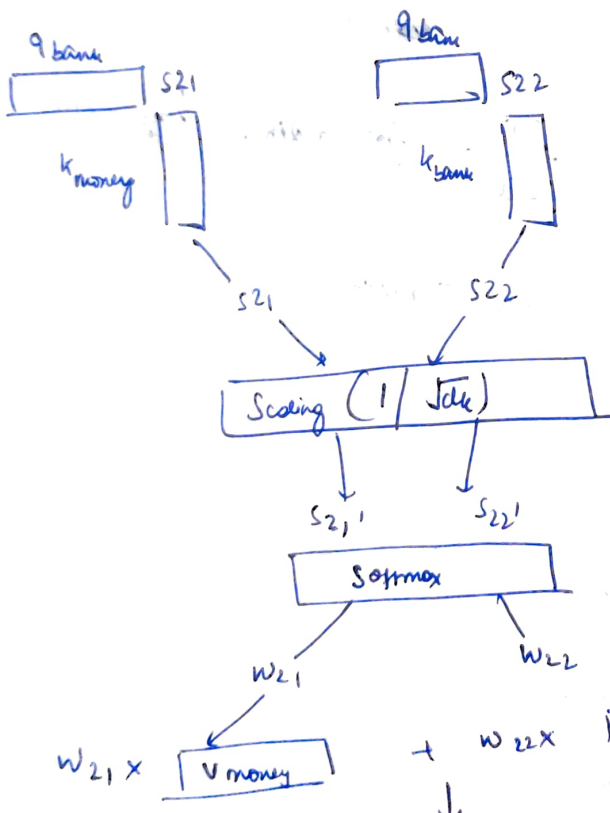


money bank



$$W_q = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad W_k = \begin{bmatrix} 3 & 4 \\ 5 & 1 \end{bmatrix} \quad W_v = \begin{bmatrix} 4 & 1 \\ 2 & 1 \end{bmatrix}$$

vector
matrix $\rightarrow$ linear transformation



Dot product

$$S_{21} = 10 \quad / \quad S_{22} = 32$$

$q_{bank} \cdot k_{money}$ $q_{bank} \cdot k_{bank}$

Scaling

$$S_{21}' = \frac{10}{\sqrt{2}} = 7.07$$

$$S_{22}' = 22.69$$

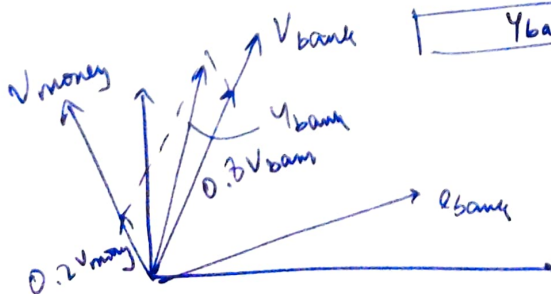
softmax

$$w_{21} = 0.2$$

$$w_{22} = 0.3$$

then

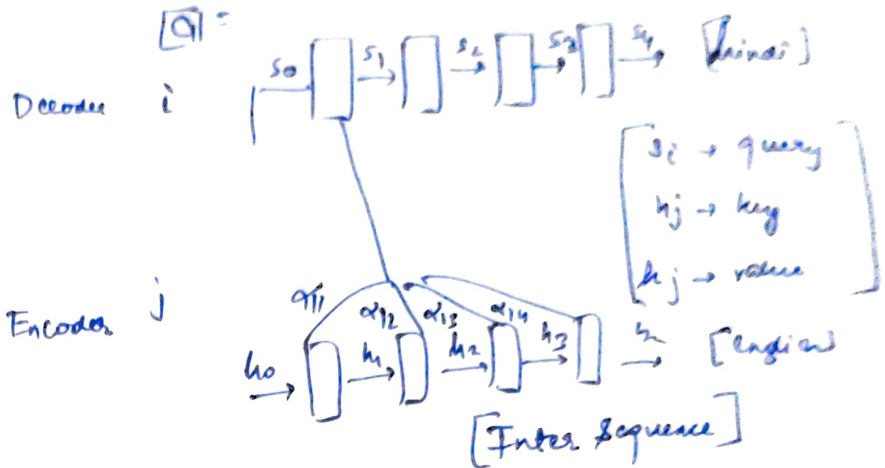
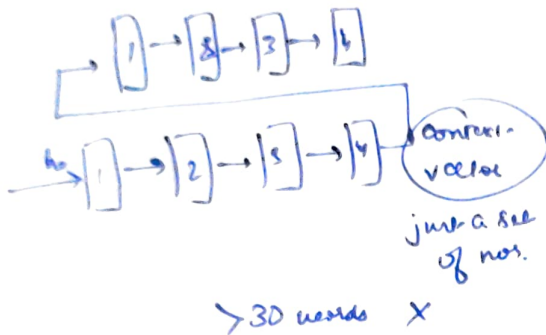
$$w_{21} \times v_{money} + w_{22} \times v_{bank} = y_{bank}$$



Why by Attention is called "Self")

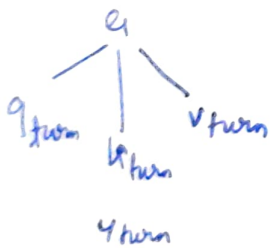
Attention mechanism

Recap of Attention



Self attention

Turn	off	the	lights
q_1	e_2	e_3	e_4
q_2	e_2	e_3	e_4
q_3	e_2	e_3	e_4



$$c_i = \sum \alpha_{ij} h_j$$

$$c_i = \alpha_{i1} h_1 + \alpha_{i2} h_2 + \alpha_{i3} h_3 + \alpha_{i4} h_4$$

$$\alpha_{ij} = \text{softmax} [z_{ij}]$$

Alignment score

$$z_{ij} = s_i^T h_j$$

$$z_{11} = s_1 h_1$$

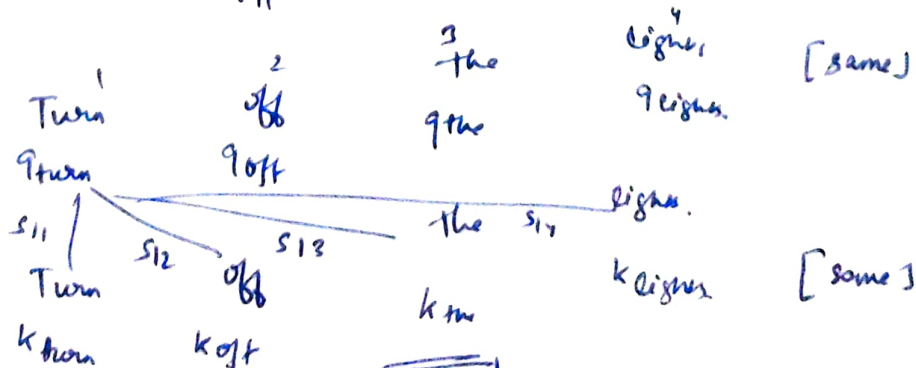
$$z_{12} = s_1 h_2$$

$$z_{13} = s_1 h_3$$

$$y_{turn} = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) \times V$$

$$y_{turn} = w_{11} v_{turn} + w_{12} v_{off} + w_{13} v_{the} + w_{14} v_{lights}$$

$$w_{11} = \text{softmax} (s_{11})$$



[Intra Sequence] $\rightarrow$ [Self]

What is Multi-head Attention in Transformers

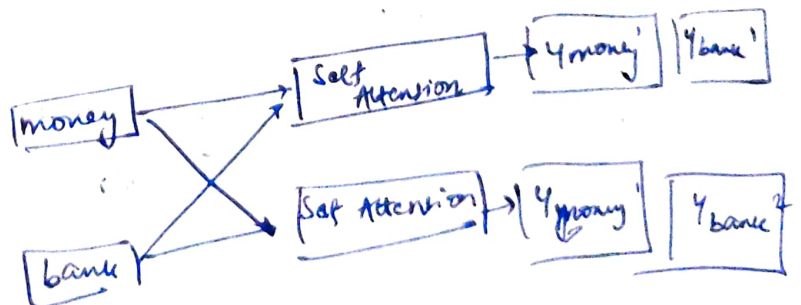
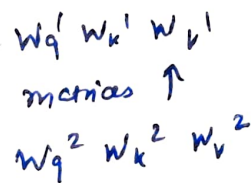
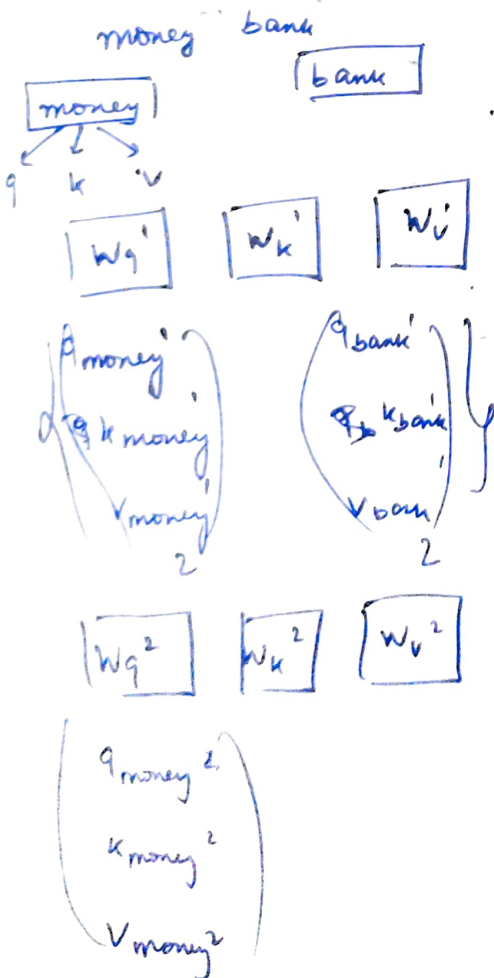
man saw the astronomer with a telescope

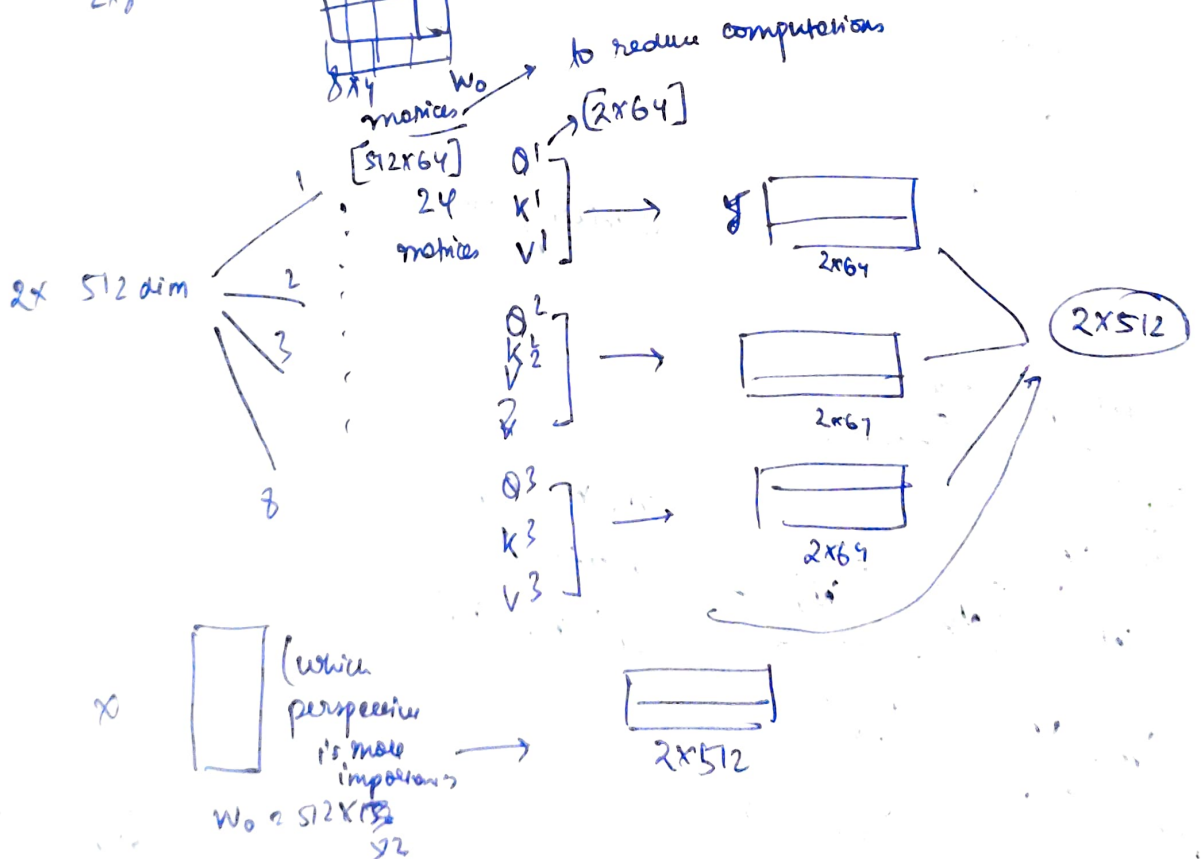
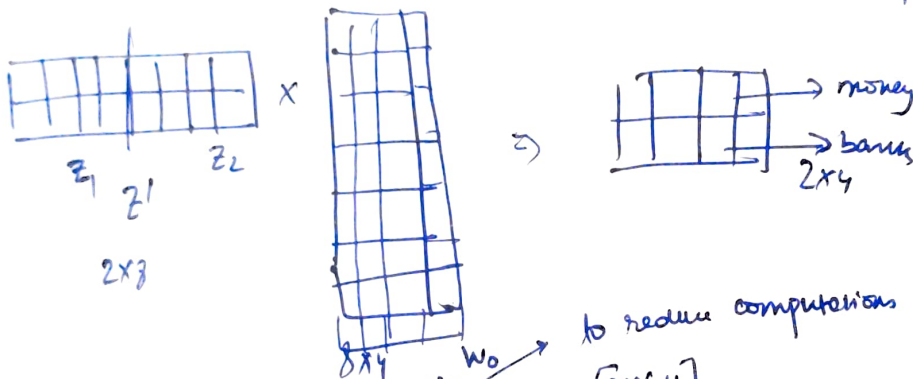
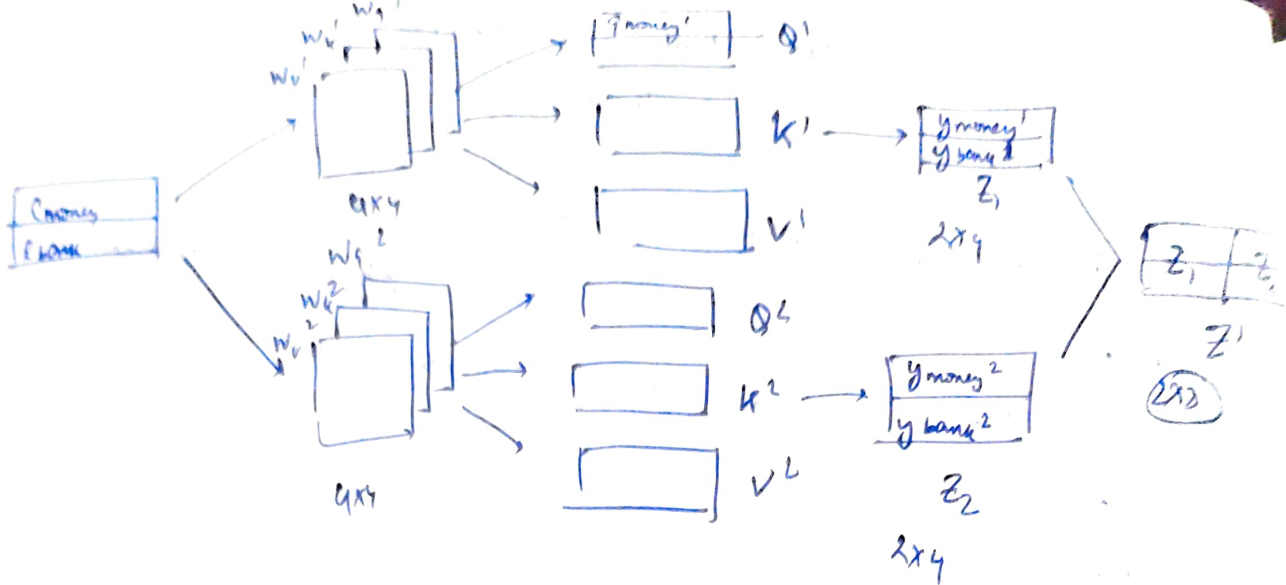
↳ obj → single perspective.

The man saw the astronomer with a telescope

The
man
saw
the
anthropomorph
with
a
telescope.

similarity table



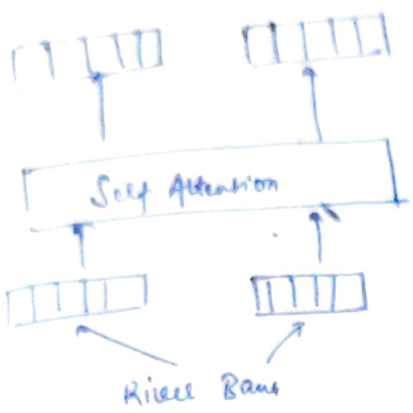


Self attention

512

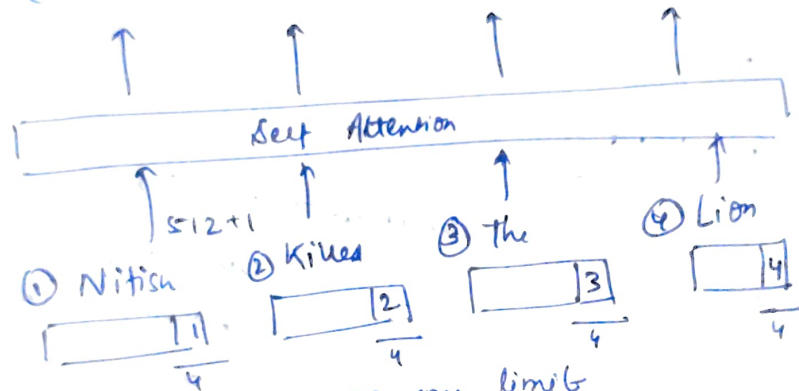
multi attention

$$512/8 = 64$$



Benefit: parallel → drawback: words order is not taken care of

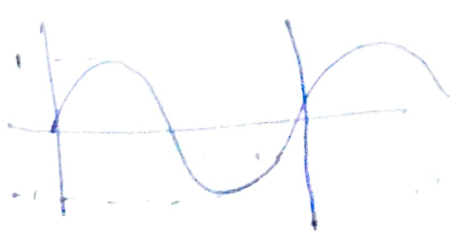
Simple Solⁿ (Doesn't work well for NN)



Problem 1) Unbounded → no upper limit

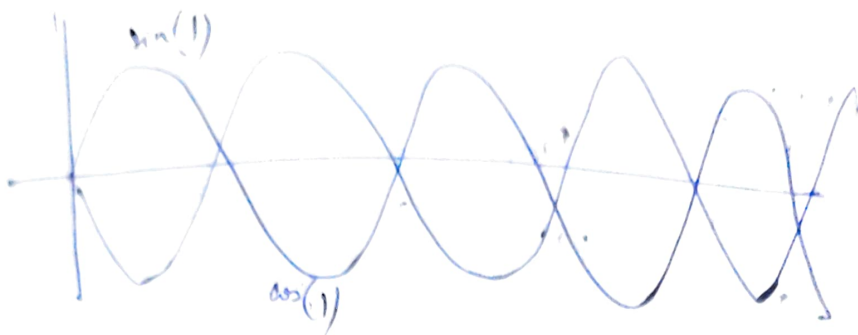
Then you
Nitish killed the lion
 $\frac{2}{4} = 0.5$ $\frac{4}{4} = 1$

- 2) 1, 2, 3, 4 → Discrete
 - 3) Can't capture relative positioning as discrete function
- absolute pos / relative position



- 1) Unbounded x
 - 2) Discrete → continuous
 - 3) Can capture relative position
- $y = \sin(1) = 0.84$
 $y = \sin(2) = 0.91$
 $y = \sin(3) = 0.14$
 $y = \sin(4) = -0.76$

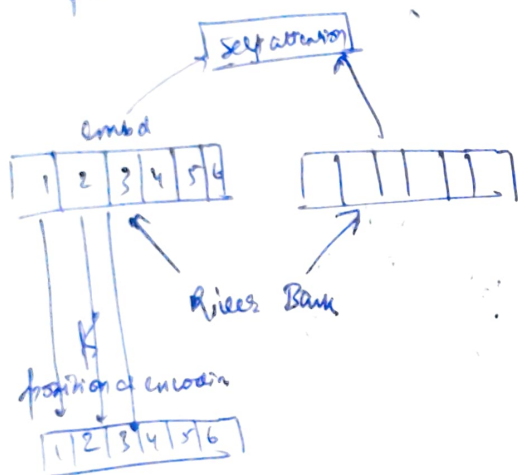
Problem → Not unique as periodic



Nitish

vector representation

positional encoding will be a vector



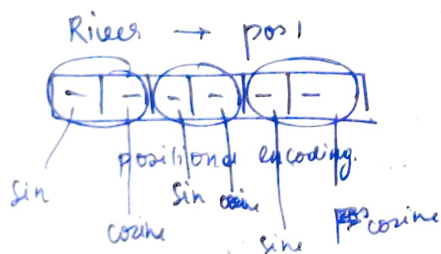
Concentration $\rightarrow$ 12 dim

$\Rightarrow$ Calculations

$\leftarrow$ parameters increased

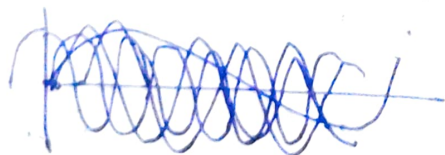
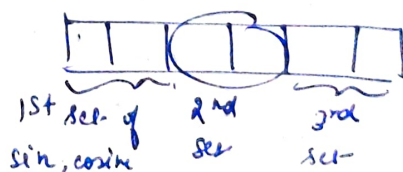
$\rightarrow$ training time will increase

Embed +
pos encoding



sin
cosine

Bank $\rightarrow$ ②



$f \downarrow$ with
increasing sine / cosine waves.

How to decide frequency?

$\rightarrow$

$$PE(pos, 2i) = \sin(pos / 10000^{2i/d_{model}})$$

$$PE(pos, 2i+1) = \cos(pos / 10000^{2i/d_{model}})$$

pos → position

pos = 0, 1, 2

dmodel → dim. of embedding

dmodel = 6

$$i = 0 - \frac{dmodel}{2} \rightarrow \underline{0-2} \quad \boxed{i=0,1,2}$$

pos=0

River

[]

pos=1

Bank

[]

[0] [] [] [] [] []

[] [] [] [] [] []

for i=0 → pos=0

$$PE(0,0) = \sin(0/10000^\circ) = 0$$

$$PE(0,1) = \cos(0/10000^\circ) = 1$$

for i=1, pos=0

$$PE(0,2) = \sin(0/10000^\circ) = 0$$

$$PE(0,3) = \cos(0/10000^\circ) = 1$$

for i=2, pos=0

$$PE(0,4) = 0$$

$$PE(0,5) = 1$$

(Bank)

for i=0, pos=1

$$PE(1,0) = \sin(1/10000^\circ) = 0.84$$

$$PE(1,1) = \cos(1/10000^\circ) = 0.54$$

i=1, pos=1

$$PE(1,2) = \sin(1/10000^\circ) = 0.04$$

$$PE(1,3) = \cos(1/10000^\circ) = 0.99$$

i=2, pos=2

$$PE(2,4) = \sin(1/10000^\circ) = 0.00$$

$$PE(2,5) = \cos(1/10000^\circ) = 0.99$$

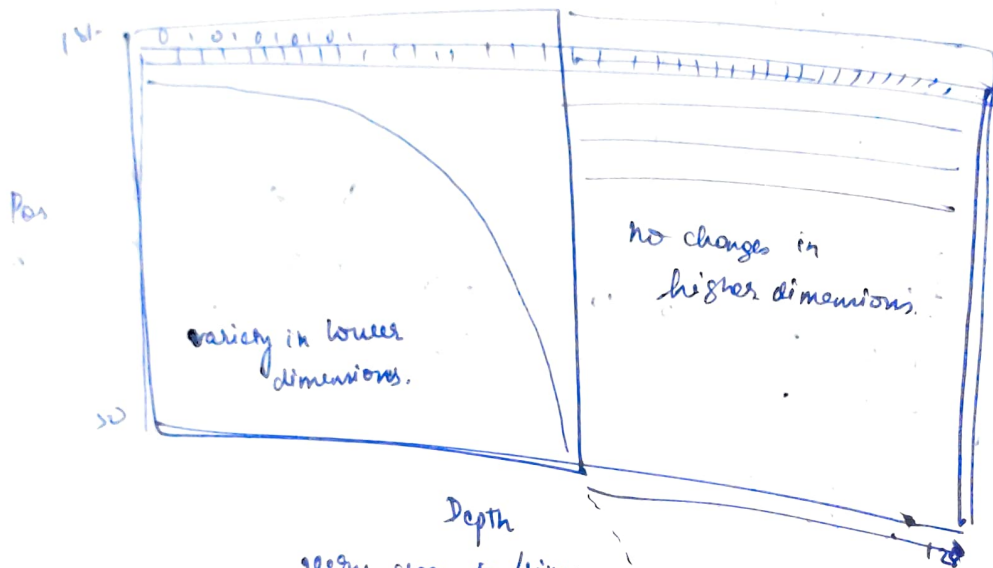
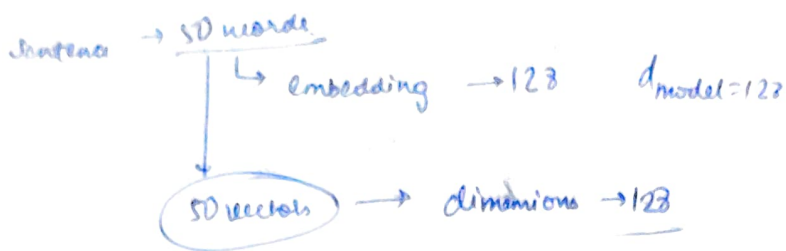
River →

[0] [1] [0] [1] [0] [1]

Bank →

[0.84] [0.54] [0.04] [0.99] [0.00] [0.99]

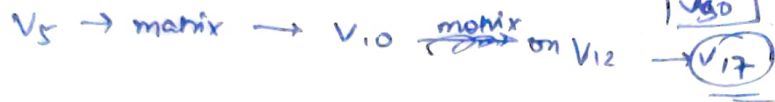
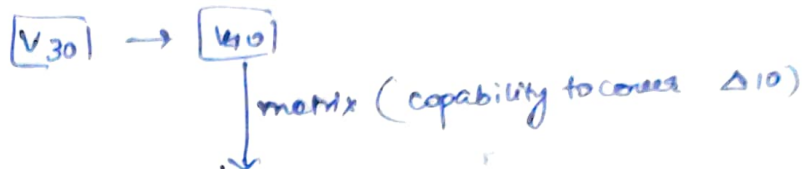
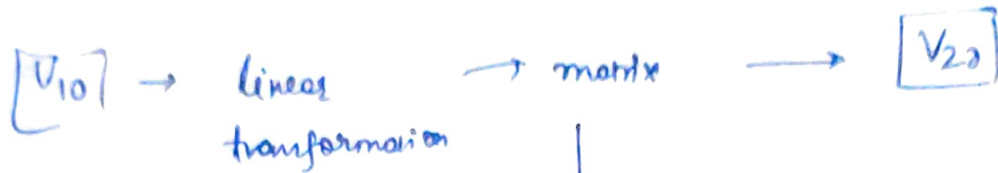
Interesting Observations



Depth
very close to binary encoding

in the domain of continuous nos.

0	:	0000
1	:	0001
2	:	0010
3	:	0011
4	:	0100
5	:	0101
6	:	0110
7	:	0111
8	:	1000
9	:	1001
10	:	1010
11	:	1011
12	:	1100
13	:	1101
14	:	1110
15	:	1111



for each Δ , one linear transformation is available.